

REPORT

Abstract: -

This project explores the impact of different prompt engineering strategies and decoding parameters on the results of two natural language processing tasks: one-sentence summarization and reading-comprehension QA.

Five prompt templates were evaluated: a baseline, role-based, reasoning-inspired, refusal-aware summarization, and strict-JSON question-answering prompt. The effect of two temperatures, 0.0 and 0.3, and a fixed top-p of 0.9 were evaluated. Due to a limited budget of Hugging Face inference credits during the full experiment, a MOCK fallback was used to complete the pipeline consistently.

The results were evaluated using lexical overlap for summarization and Exact Match (EM), F1, execution success rate, and JSON validity for QA. This experiment provides a reproducible evaluation pipeline while demonstrating the effects of different prompting strategies, decoding parameters, output constraints, and infrastructure limitations.

Methods: -

The experiment looked at the effects of various prompt-engineering approaches on two tasks: one-sentence summarization and reading-comprehension question answering. The generated dataset contained 120 examples in total, of which 70 examples were summarization and 50 examples were QA. We devised five prompt templates in total: baseline summarization, role-based summarization, reasoning-inspired summarization, refusal-aware summarization, and strict-JSON QA. The QA prompt required that the response contain an answer and supporting evidence in valid JSON format.

We evaluated the effectiveness of two decoding temperatures, 0.0 and 0.3, while keeping top-p at 0.9 and the maximum generation length at 196 tokens. The notebook was set up to use the open-source Qwen/Qwen3.8-27B model from the Hugging Face model hub via their inference API. The integration with Hugging Face was tested, but the available number of inference credits was exhausted before the full experimental grid could be run. Therefore, the notebook's allowed MOCK fallback was used in the final complete run.

For the final MOCK experiment, we evaluated four summarization prompts and two temperatures for a total of 320 summarization generations and one QA prompt across two temperatures for 80 QA generations. In total, a MOCK experiment of 400 generations was performed. Summarization was evaluated using lexical overlap, while QA was evaluated using Exact Match and token-level F1 after extracting the answer field from the JSON response. We also computed the execution success rate and valid-JSON rate.

Results: -

The final full set of experimental runs resulted in 400 responses, all of which executed successfully (execution success rate: 100%), with no execution errors. For the purposes of summarization, the baseline, role-based, reasoning-inspired, and refusal-aware prompt strategies each had a mean lexical-overlap score of approximately 0.8104 in the MOCK evaluation. The results were the same at temperatures 0.0 and 0.3, as would be expected, since the MOCK implementation is entirely deterministic and does not perform any probabilistic generation.

On the QA task, the strict-JSON prompt achieved an execution success rate of 100% and a valid-JSON rate of 100% at both temperatures. However, when the answer field was extracted from each JSON-formatted response and compared against the reference answer, the resulting Exact Match score was 0.125 and the F1 score was 0.125 at both temperatures. Thus, despite the high execution and formatting success rates, the model mostly failed to provide the correct answer.

Since the final full set of experiments required a MOCK fallback after the Hugging Face inference credits were exhausted, these quantitative results should be primarily interpreted as a validation of the experiment, evaluation, and visualization pipeline, and not as a demonstration of the capabilities of the real Qwen model.

Error Analysis & Hallucination Audit: -

I have identified two types of limitations in my experiment: infrastructure shortcomings and response quality shortcomings. My attempts to utilize the Hugging Face infrastructure resulted in the HTTP 402 error due to reaching the limit of free credits. Thus, the experiment grid could not be completed with the real model; instead, I had to rely on the MOCK implementation of the QA API.

Even though the MOCK implementation had a 100% QA execution success rate and 100% JSON validity, these metrics do not necessarily reflect the correctness of answers. According to the corrected QA results, the MOCK version achieved EM = 0.125 with F1 = 0.125. In many cases, the MOCK implementation returned CANNOT_ANSWER when the context contained the answer, so I believe that these are examples of incorrect refusal to answer rather than hallucinations.

As for the summarization task, the average lexical overlap was only 0.8104 across all four prompt variations. However, I think that the results cannot be used to demonstrate the prompt sensitivity of the model since the same prompt and temperature produced identical results in the MOCK implementation. Thus, it would be unwise to draw any further conclusions about hallucinations or comparative prompt quality before testing the hypothesis on a real model.

Discussion: -

The experiment is illustrative of evaluating prompt engineering from the perspectives of prompt variations, task specifics, and decoding parameters. The summarization instructions were exemplary of different approaches, such as a basic prompt, role prompting, reasoning prompting, and refusal prompting. The strict-JSON QA prompt is illustrative of how the requirements could be specified to make the parsing and automated evaluation easier.

The variation of the temperature parameter between 0.0 and 0.3 made no difference in the final run's lexical overlap, EM, or F1. This finding must not be interpreted as a negative statement about the effect of temperature in real LLMs. MOCK implementation is deterministic, so changing the temperature does not affect decoding. The Hugging Face integration is used to show that the notebook was set up to work with the open-source model, but due to the limited inference credits, the actual model's experimental grid could not be completed. Thus, the final quantitative results mostly showcase the whole prompt engineering, generation, evaluation, error-checking, and visualization pipeline.

Conclusion: -

This project created an end-to-end framework for evaluating prompt engineering strategies on summarization and reading-comprehension QA. Five prompt templates are investigated using two temperature settings, and the results are compared based on lexical overlap, Exact Match, F1, execution success rate, and JSON validity. CSV artifacts and visualization are also produced for enhanced analysis.

While the final complete experiment used the MOCK fallback due to the Hugging Face inference credit limit, the attached notebook demonstrates the full process from prompt construction, generation, evaluation, error analysis, and visualization. One key observation is that many successful runs (high success rate and valid JSON) have rather mediocre quality responses. It is also important to note that all these results were obtained using a single model; therefore, future work should re-run the complete experiment to have enough inference credits to use different open-sourced models and compare their prompt effectiveness, temperature sensitivity, robustness, diversity of responses, and tendency to hallucinate.

Ethics & Limitations: -

Prompt-engineered LLMs can generate wrong, incomplete, biased, or unsound information even if it follows the requested format. It is especially crucial for QA and summarization systems, as fluent or correctly structured but semantically wrong answers could mislead users into trusting incorrect information. Requiring evidence, requiring strict output formats like JSON, refusing to answer when evidence is not found, and human evaluation can alleviate these issues.

The main limitation of this experiment was the Hugging Face inference-credit allotment, which did not allow completing the experimental grid with the real Qwen model. Hence, the final complete evaluation grid was based on a MOCK implementation, which has deterministic outputs and cannot replicate the whole range of behaviors of a real generative model. As such, the prompt engineering efficacy, temperature sensitivity, hallucination, robustness, and response diversity cannot be evaluated. The lexical overlap, EM, and F1 are also only approximate measurements of response quality and do not necessarily reflect factual correctness or semantic similarity. Future experiments should involve sufficient real-model inference, other models, semantic similarity evaluation, and human evaluation.

Takeaways: -

The conducted experiment generates several valuable insights concerning the design and assessment of the LLM-based solutions:

1. Prompt structure has proven to be important: clear task instructions, role definitions, refusal rules, and structured-output requirements can aid in controlling and assessing LLM responses.
2. Multiple evaluation metrics are necessary: execution success alone does not demonstrate quality. Lexical overlap, Exact Match, F1, and output-format validity serve as complementary assessment vectors.
3. Structured outputs have proven to improve usability: requiring QA responses in a valid JSON format makes them easier to parse and process downstream, but a valid format does not imply correctness.
4. Decoding parameters should be evaluated systematically: temperature can impact real-model consistency and diversity, but no meaningful temperature impact could be assessed using the deterministic MOCK implementation.
5. Infrastructure affects reproducibility: API availability, inference quotas, and credit consumption must be considered when designing LLM experiments. The Hugging Face credit limitation in this experiment is a lesson about error handling, fallback mechanisms, and transparent reporting of infrastructure limitations.